Lezione 10 01/04/26 3 ore

- Variabili discrete

$X \in \{x_1, \dots, x_n, \dots\}$ $\quad p_1 = \mathbb{P}[X_1] \dots, p_n = \mathbb{P}[X = x_n], \dots$

$p_n \geq 0 \quad \forall n \geq 1$

$p_1 + \dots + p_n + \dots = \underbrace{\sum_{n=1}^{+\infty} p_n}_{\text{serie numerica}} = 1$

$\sum_{n=1}^{+\infty} p_n = \underbrace{\lim_{n \to +\infty}}_{\text{limite}} \underbrace{(p_1 + \dots + p_n)}_{\text{somma finita}} =$

Esempio Lancio una moneta finché non esce testa $\mathbb{P}[T] = \frac{1}{2}$

N = numero di lanci $\quad N \in \{1, 2, \dots, n, \dots\} = \mathbb{N} - \{0\}$

$\mathbb{P}[N = n] = \underbrace{\left(\frac{1}{2}\right)^{n-1}}_{(n-1) \text{ croci}} \underbrace{\frac{1}{2}}_{1 \text{ testa}}$

$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} + \dots = (\text{in base binaria})$

T $\quad$ CT $\quad$ CCT $\quad$ $\underbrace{C \dots C}_{n-1} T$

$= (0,1)_2 + (0,01)_2 + (0,001)_2 + \dots + (0, \underbrace{0 \dots 0}_{n-1} 1)_2 + \dots$

$= (0, 1\,1 \dots 1 \dots)_2 = (0, \bar{1})_2 = 1$

- Variabile con densità (o assolutamente continua)

Dati $a, b \in \mathbb{R}$ $\quad a < b$

$$\mathbb{P}[X \in [a,b]] = \mathbb{P}[a \le X \le b] = \int_a^b f(x)\,dx$$

dove $f : I \to \mathbb{R}$ con dominio $I = \mathbb{R} \setminus \{x_1, \dots, x_n\}$

è continua e integrabile in senso improprio

in $x_1, \dots, x_n$ e in $-\infty$ e $+\infty$

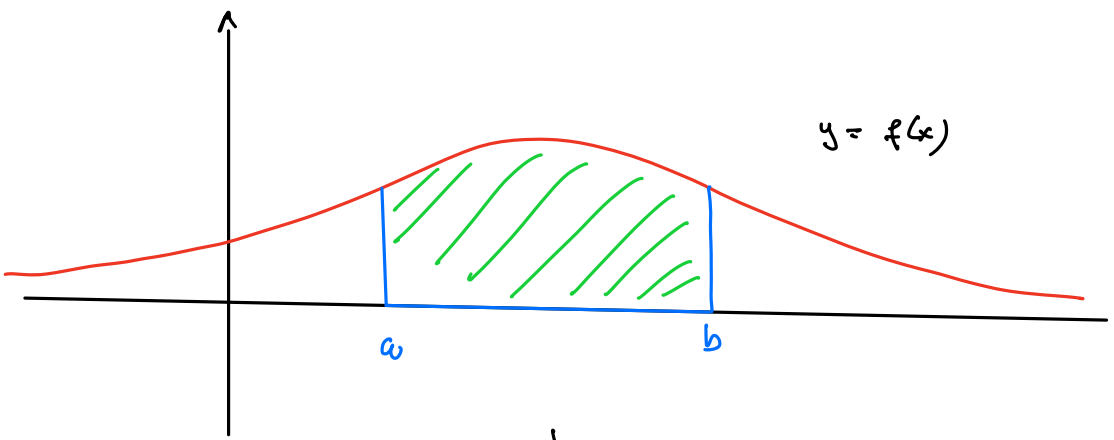
la funzione f è detta densità di probabilità di X

$I = (-\infty, x_1) \cup (x_2, x_3) \cup \dots \cup (x_{n-1}, x_n) \cup (x_n, +\infty)$



- Esempio

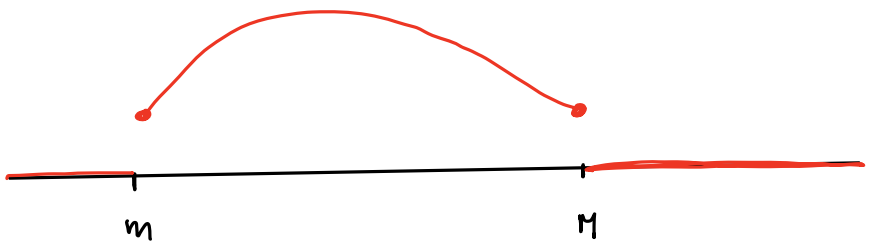
se $f : \mathbb{R} \to \mathbb{R}$ continua $\quad I = \mathbb{R}$



$$\mathbb{P}[a \le X \le b] = \int_a^b f(x)\,dx = \text{area regione verde}$$

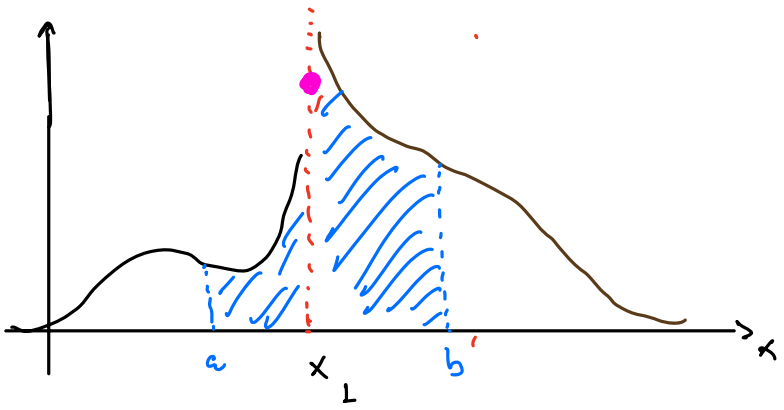
- $m \le X \le M \iff X \in [m, M] \iff$ $\quad f(x) = 0$ se $x < m$

$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\quad f(x) = 0$ se $x > M$



- Esempio

$f : \mathbb{R} \setminus \{x_1\} \to \mathbb{R}$ $\qquad I = (-\infty, x_1) \cup (x_1, +\infty)$



$y = f(x)$

$a < x_1 < b$

$$\mathbb{P}[a \le X \le b] = \int_a^{x_1} f(x)\,dx + \int_{x_1}^{b} f(x)\,dx =$$

$$= \lim_{x \to x_1^-} \int_a^x f(x)\,dx + \lim_{x \to x_1^+} \int_x^b f(x)\,dx$$

$$\mathbb{P}[X \ge b] = \int_b^{+\infty} f(x)\,dx = \lim_{c \to +\infty} \int_b^c f(x)\,dx$$

Oss La densità f soddisfa le seguenti proprietà

i) $f(x) \ge 0 \qquad \forall x \in \mathbb{R}$

ii) $\int_{-\infty}^{+\infty} f(x)\,dx = 1$

Prop Se X è una variabile aleatoria con densità f allora

a) $\mathbb{P}[X = x] = 0 \quad \forall x \in \mathbb{R}$

b) $\mathbb{P}[a \le X \le b] = \mathbb{P}[a < X < b]$

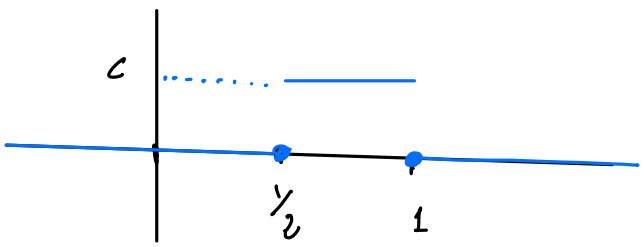
$\mathbb{P}[a \le X < b] = \mathbb{P}[a < X \le b] = \int_a^b f(x)\,dx$

c) $\mathbb{P}[X \le b] = \mathbb{P}[X < b] = \int_{-\infty}^{b} f(x)\,dx$

$\mathbb{P}[X \ge a] = \mathbb{P}[X > a] = \int_a^{+\infty} f(x)\,dx$

Esempio

X numero estratto uniformemente tra $\frac{1}{2}$ e 1



$$f(x) = \begin{cases} 0 & x \le \frac{1}{2} \\ c & \frac{1}{2} < x < 1 \\ 0 & x \ge 1 \end{cases}$$

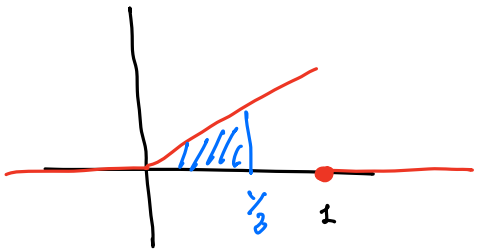
poiché l'estrazione è uniforme

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{\frac{1}{2}}^{1} c\,dx = \underbrace{\left(1 - \tfrac{1}{2}\right)}_{\text{base}} \underbrace{c}_{\text{altezza}} = \frac{c}{2} = 1 \Rightarrow c = 2$$

$$f(x) = \begin{cases} 0 & x \le \frac{1}{2} \\ 2 & \frac{1}{2} < x < 1 \\ 0 & x \ge 1 \end{cases}$$

Esempio

$$f(x) = \begin{cases} 0 & x < 0 \\ mx & 0 \le x \le 1 \\ 0 & x > 0 \end{cases}$$



$0 \le x \le 1$

$m \ge 0$

$$\int_0^1 mx\,dx = m\,\frac{x^2}{2}\Big|_0^1 = \frac{m}{2} = 1 \qquad m = 2$$

$$P[\,X \le 0\,] = 0$$

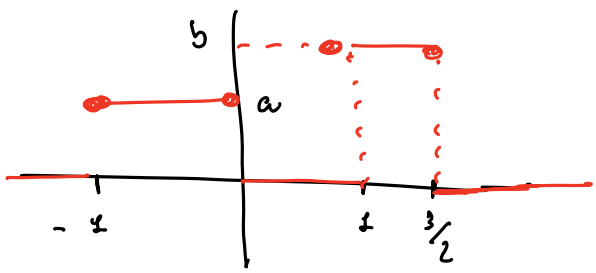
$$P\left[\,|x| \ge \tfrac{1}{2}\,\right] = P\left[X \le -\tfrac{1}{2}\right] + P\left[X \ge \tfrac{1}{2}\right] = 0 + 1 - P\left[X \le \tfrac{1}{2}\right]$$

$$= 1 - \frac{1}{2}\underbrace{\left(\frac{1}{2} - 1\right)}_{\text{base}}\underbrace{2\,\frac{1}{2}}_{\text{altezza}} = 1 - \frac{1}{4} = \frac{3}{4}$$

$$= \int_{1/2}^{1} 2x\,dx = x^2\Big|_{1/2}^{1} = 1 - \frac{1}{4} = \frac{3}{4}$$

Esempio

$$f(x) = \begin{cases} a & -1 \le x \le 0 \\ b & 1 \le x \le 3/2 \\ 0 & \text{altrove} \end{cases}$$



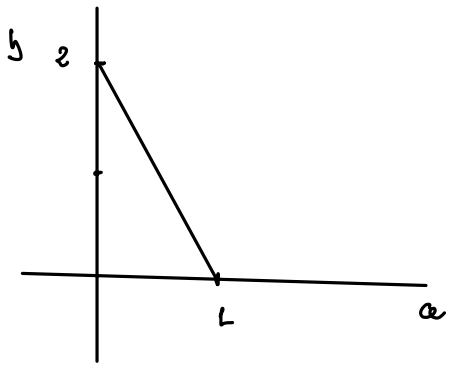
$a \ge 0 \qquad b \ge 0$

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-1}^{0} a\,dx + \int_{1}^{3/2} b\,dx = 1$$

$$= a + \frac{b}{2} = 1$$

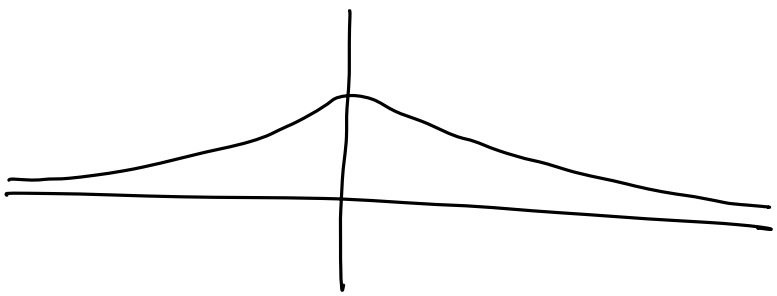
$$\begin{cases} b = 2 - 2a \\ a \geq 0 \qquad 2 - 2a \geq 0 \end{cases}$$

$$\begin{cases} b = 2 - 2a \\ 0 \leq a \leq 1 \end{cases}$$



Esempio

$$f(x) = \frac{a}{1+x^2}$$



$a \geq 0$

$$\int_{-\infty}^{+\infty} f(x)\,dx = a\int_{-\infty}^{0} \frac{1}{1+x^2}\,dx + a\int_{0}^{+\infty} \frac{1}{1+x^2}\,dx$$

$f(-x) = f(x)$

$$= 2a\int_{0}^{+\infty} \frac{1}{1+x^2}\,dx = 2a \lim_{b\to+\infty} \int_{0}^{b} \frac{1}{1+x^2}\,dx$$

$$= 2a \lim_{b\to+\infty} \operatorname{arctg} x \Big|_0^b = 2a \lim_{b\to+\infty} \operatorname{arctg} x = a\pi$$

$$\Rightarrow \quad a = \frac{1}{\pi}$$